Design and TCAD Simulation of MoS_2-Based FeFETs with HfO_2 Gate Oxide for Circuit Applications

ASGAR ALI
M.Tech VLSI DESIGN
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Department of Electrical Engineering
IIT Jammu
Under the Guidance of:
Dr.Chandan Yadav

Table of Contents

1. Previous Work
2. Literature Review of Current Work
3. TCAD implemented FeFET
4. Device Simulation and Results
5. Content-Addressable Memory (CAM) Cell
6. Search Operation Mechanics
7. Future Work
8. References

Previous Work – MoS2 Memtransistor

- **Objective:** Implement an MoS_2 channel memtransistor using an HfO_2 dielectric.
- **Proposed Mechanism:** Modulation of threshold voltage (V_{th}) via charge trapping and detrapping at the dielectric interface.
- **Operation:** Application of negative pulses for programming (detrapping) and positive pulses for erasing (trapping)
- **Methodology:**Device structure created in SDE; DC transfer characteristics ($I_d - V_g$) obtained in Sdevice.

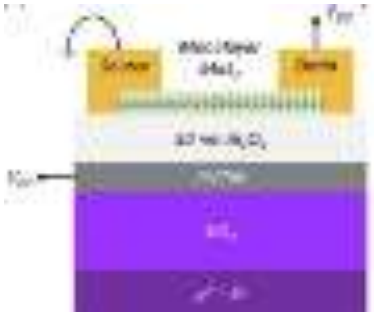


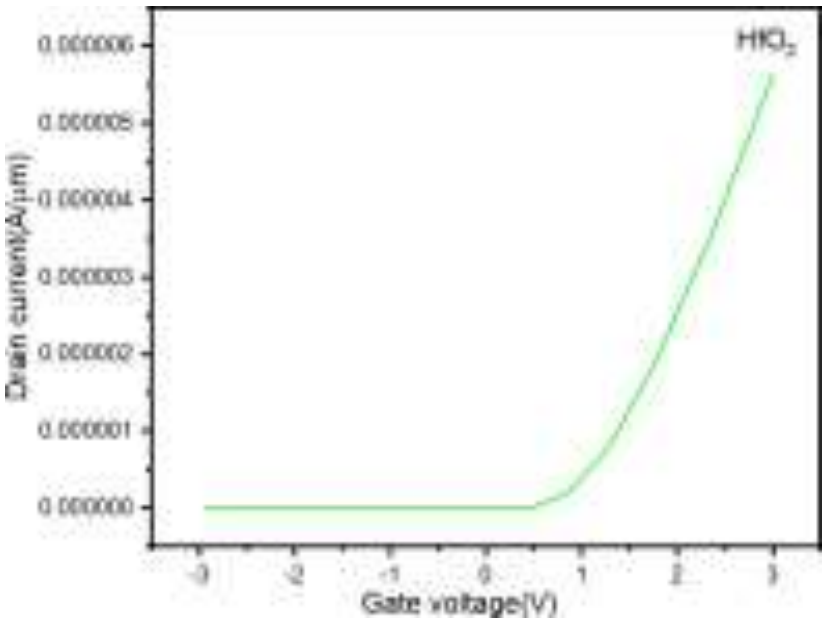
Figure: 2D Back gated memtransistor[1]

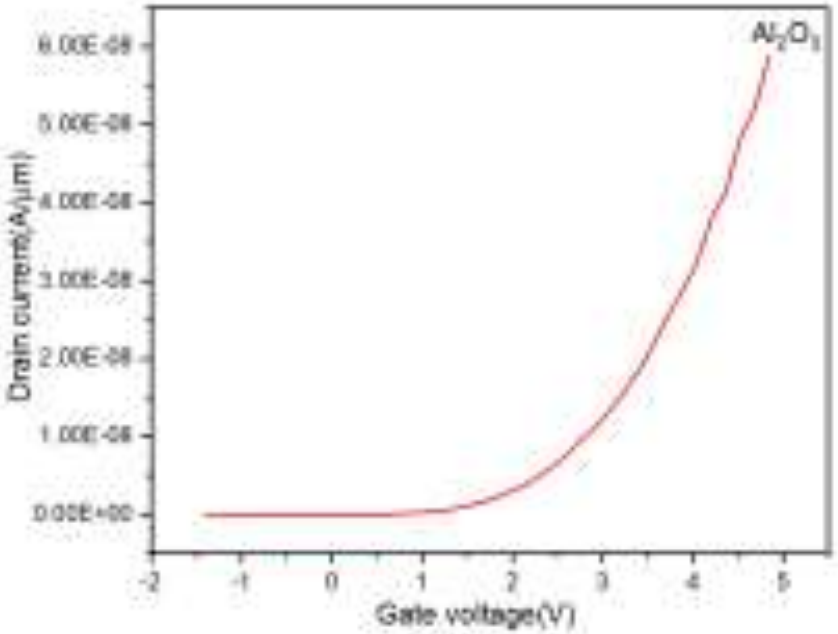
Previous work simulation

Simulation methodology

- **Simulation Limitations:** While DC transfer characteristics ($I_d - V_g$) were successfully obtained, replicating dynamic V_{th} shifts through pulse application proved computationally elusive in TCAD..
- **Modeling Complexity:** The failure was caused by the difficulty of mimicking stochastic trapping behavior—governed by SRH dynamics and energetic trap states (E_{it})—under short-duration pulses.
- **Quasi-Static Workaround:** Shifted to a quasi-static approach, manually adjusting interface trap concentrations during DC simulations to analyze the resulting impact on current and V_{th} shifts.

Simulation results





Previous work simulation

Interface trap concentrations(Dit) and interface traps energy level(Eit) variations

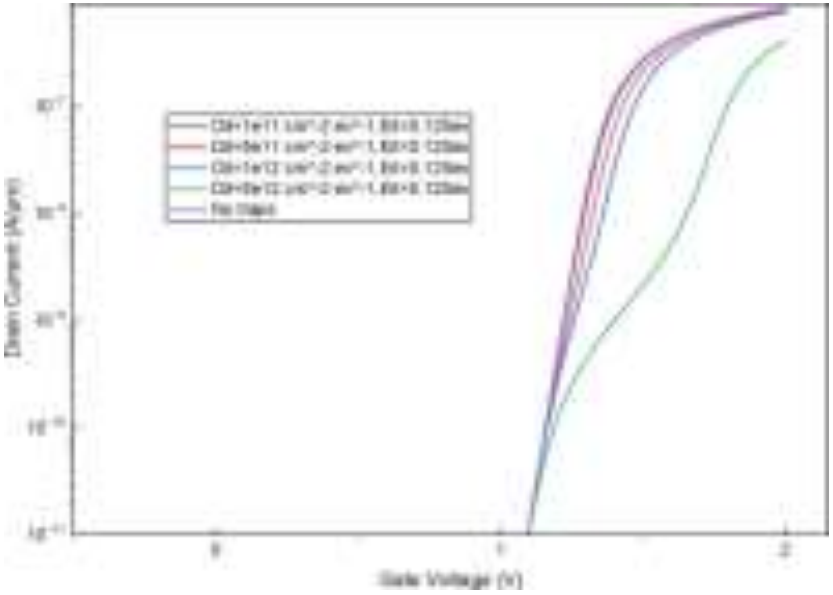


Figure: Ids vs Vgs Different Dit at 0.125eV

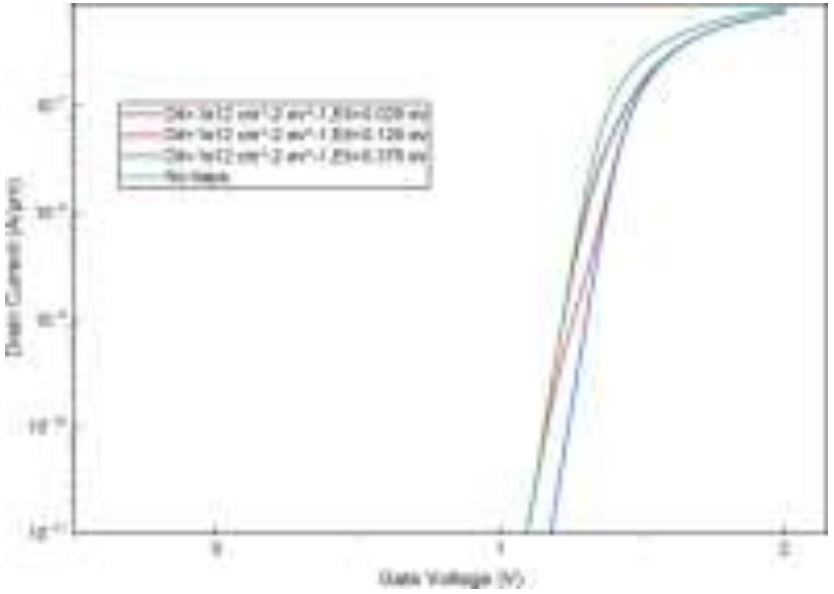


Figure: Id vs Vgs at different Eit while Dit is fixed(1e12/cm^2.eV^1)

Literature Review of Current Work

Jianjian Wang et al., "Effects of Charge Trapping on Memory Characteristics for HfO_2-Based Ferroelectric Field Effect Transistors", Nanomaterials (2023).[2]

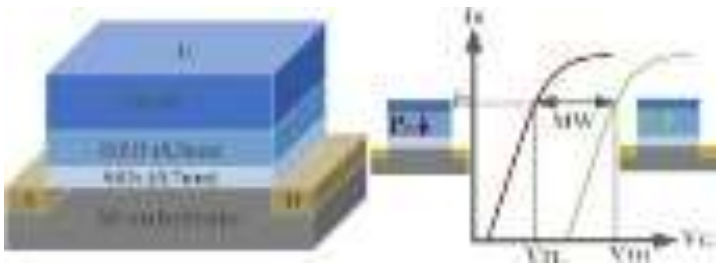


Figure: Schematic structure ferroelectric field effect transistor (FeFET) devices and the transfer characteristic curves (ID-VG) under the opposite spontaneous polarization [2]

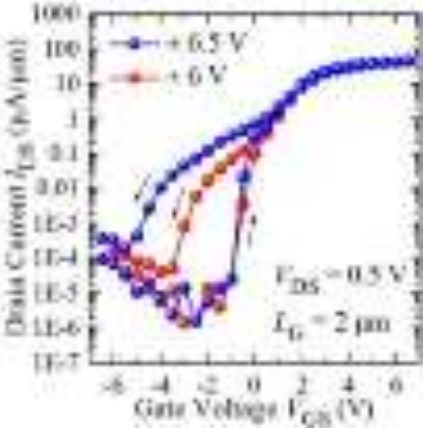


Figure: Id vs Vgs of two states of FeFET[2]

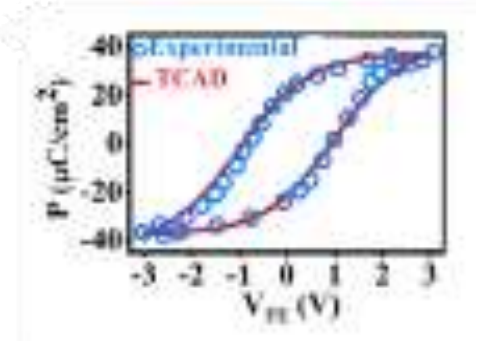


Figure: Polarization of two states of FeFET[2]

- **Polarization Mechanism:** Data is stored via polarization reversal; a positive gate voltage induces downward polarization (Low-V_T state), while a negative voltage induces upward polarization (High-V_T state).

MoS_2-Channel FeFET

Reference: Pin-Chun Shen et al., "Ferroelectric memory field-effect transistors using CVD monolayer MoS_2 as resistive switching channel", Applied Physics Letters (2020).[3]

- **Operation Physics:** Positive gate bias aligns dipoles toward the MoS_2, creating an electron accumulation regime and a Low Resistance State (LRS).

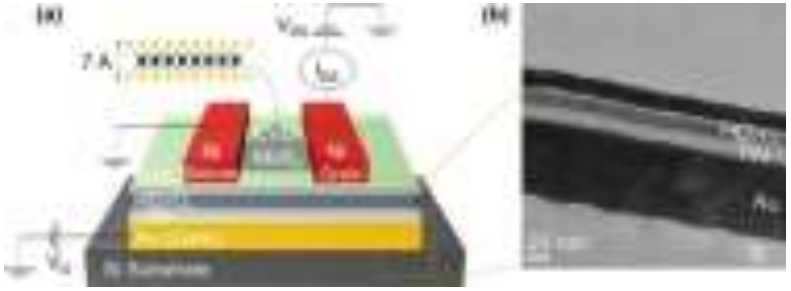


Figure: (a) Schematic diagram of a single-layer MoS2 FeFET. The gate stack is composed of an Au gate electrode, TiN, HfZrOx as the ferroelectric layer. (b) Cross-sectional view of a representative sample showing the structure and thicknesses of the gate stack.[3]

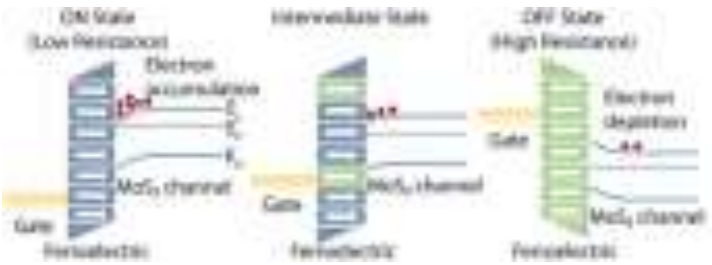


Figure: Energy band diagram of the MoS2 FeFET for different working states including a low resistive, intermediate, and high resistive states[3]

TCAD implemented FeFETs

- **Device Structure:** A MOSFET with a ferroelectric layer (e.g., HfO_2) embedded in the gate stack. .
- **Working Principle:** Stores data as the physical orientation of electric dipoles.
  - **Positive Gate Bias**: Dipoles point toward the channel, accumulating electrons ($Low - V_{th}$ state).
  - **Negative Gate Bias:**Dipoles point toward the gate, depleting the channel ($High - V_{th}$ state).

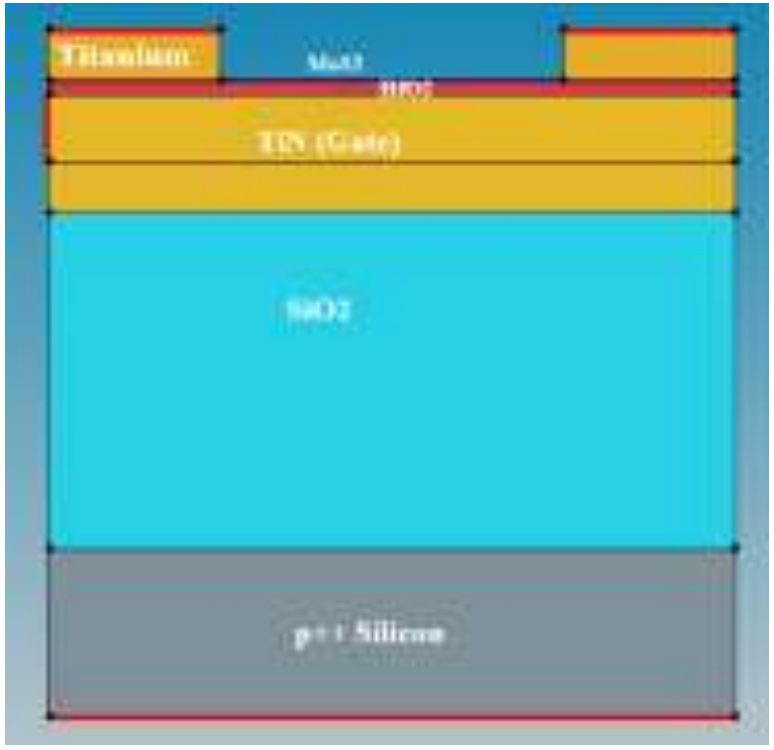


Figure: 2D Device structure sde(Structural Setup (SDE): Monolayer MoS_2 channel, 8 nm HfO_2 ferroelectric, TiN gate, and Titanium S/D contacts.)

Device Simulation Results

TCAD Simulation Methodology

- **Structural Setup (SDE):** Monolayer MoS_2 channel, 8 nm HfO_2 ferroelectric, TiN gate, and Titanium S/D contacts.
- **Physics Models (Sdevice):**
  - **Polarization:** Preisach Model for HfO2 hysteretic behavior.
    Physics (Material= "HfO2") Polarization
- **Remanent Polarization (P_r):** The polarization that remains in the HfO_2 layer after the external electric field is removed.
- **Saturation Polarization (P_s):** The maximum polarization level achieved .
- **Coercive Field (F_c):** The critical electric field intensity required to switch the dipole direction.
- **Dynamic Parameters:** Uses specific relaxation times $\tau_E = 1 \times 10^{-9}$ s and $\tau_P = 0$ s to control the temporal switching response during pulse applications

Parameter	HfO_2 Device
P_r	10–30 $\mu C/cm^2$
P_s	20–50 $\mu C/cm^2$
F_c	1–2 MV/cm
τ_E	1×10^{-9}

Electrical parameters of the HfO_2 device [2]

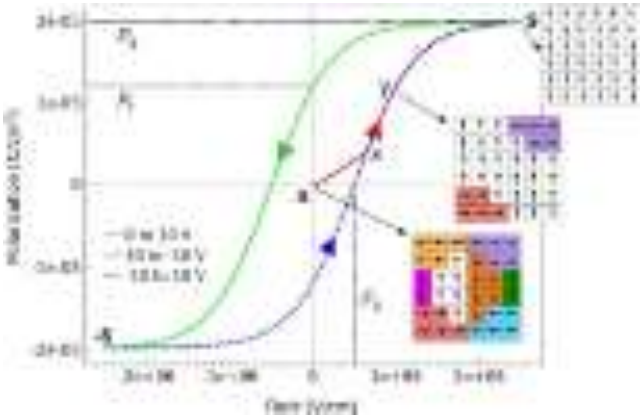


Fig:Polarization vs Gate Voltage[5]

Ferroelectric Simulation Results

IdVg characteristics for +3V(Low Vt) and -3v (High Vt) state:

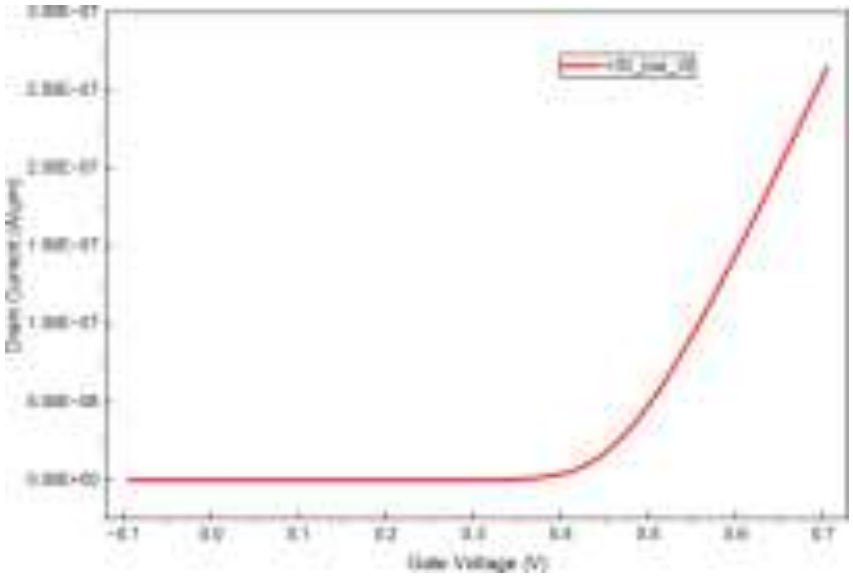


Figure: IdVg for +3v Low Vt State

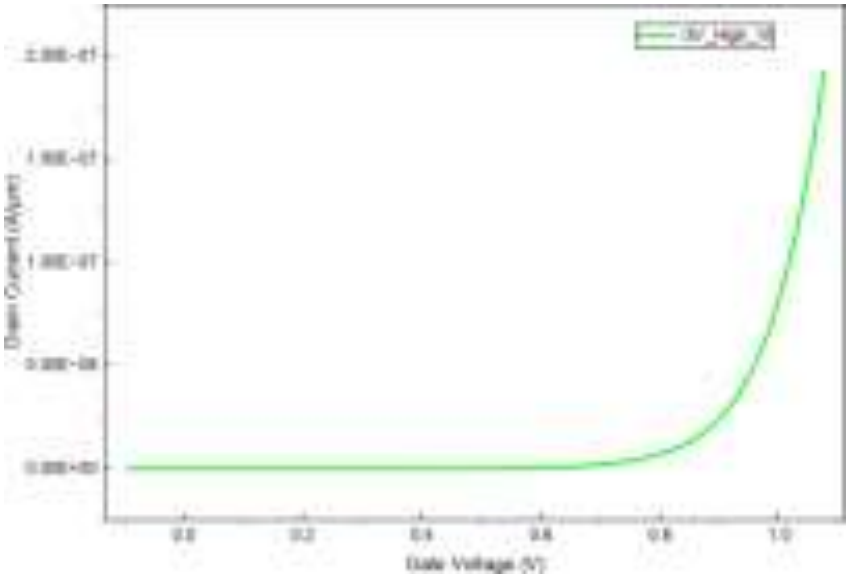


Figure: IdVg for -3v High Vt State

Vth shift and Polarization simulation results

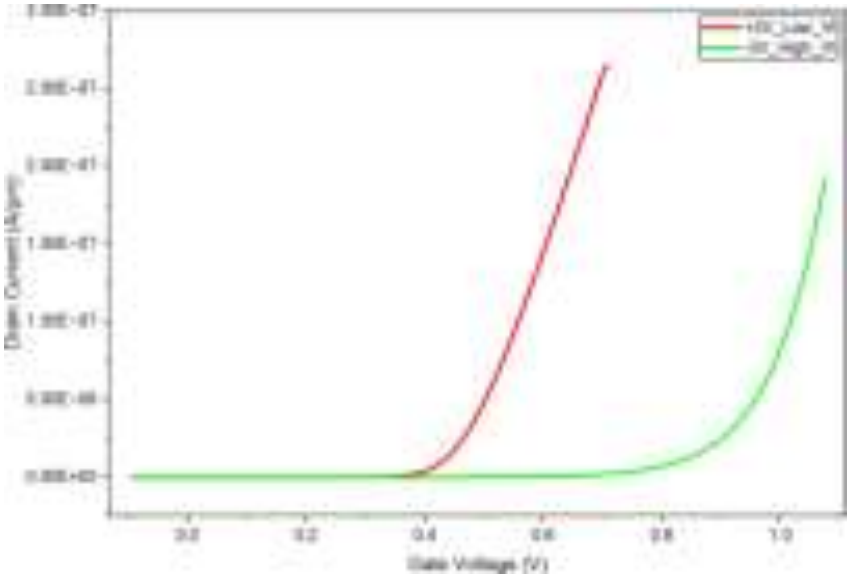


Figure: IdVg for +3v and -3v pulse state

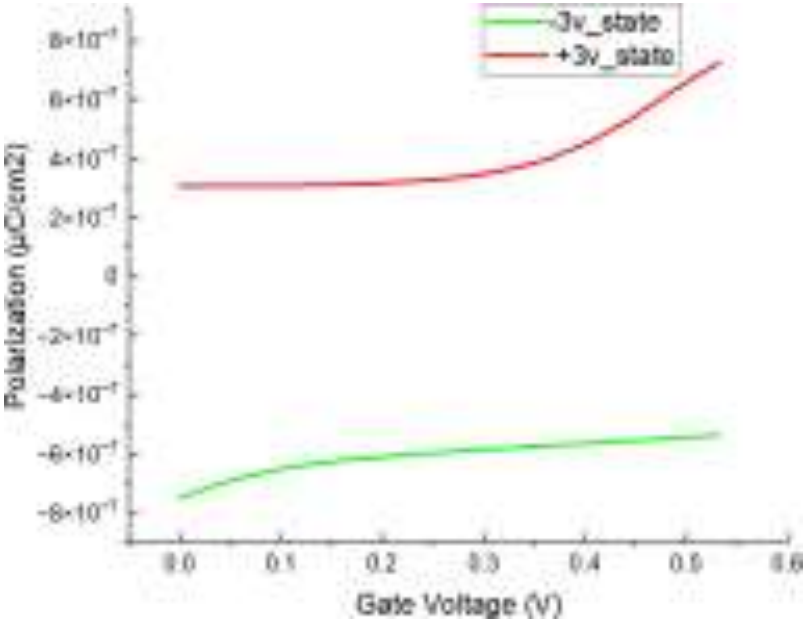


Figure: Polarization vs gate voltage for +3v and -3v states

From Device Simulation to Circuit Application

- The MoS_2-based FeFET device exhibits two stable threshold voltage states (Low Vth and High Vth) due to polarization switching.
- These two Vth states are utilized at the circuit level to implement a CAM cell, enabling storage of logic '0' and '1' using two FeFET devices.

Device Simulation

MoS_2 based FeFET Device

Device working at two V_{th}
Low V_{th} and High V_{th}

→

Circuit Simulation (Mixed Mode)

Content Addressable Memory Cell (CAM)

CAM cell Write 0 and Write 1
Using two FeFET devices

Content-Addressable Memory (CAM) Cell

Introduction to CAM Cells

- **Definition:** Provide the data and memory will tell you that if it has data and where it is .
- **Architecture:** 2-FeFET cell.
- **Key Circuit Element (R_{dd}):** A 1 MΩ pull-up resistor connected to the Match Line (ml).
- **Function of R_{dd}:** Manages Match Line pre-charge/discharge; if a mismatch conducts, R_{dd} limits current while ml voltage drop.

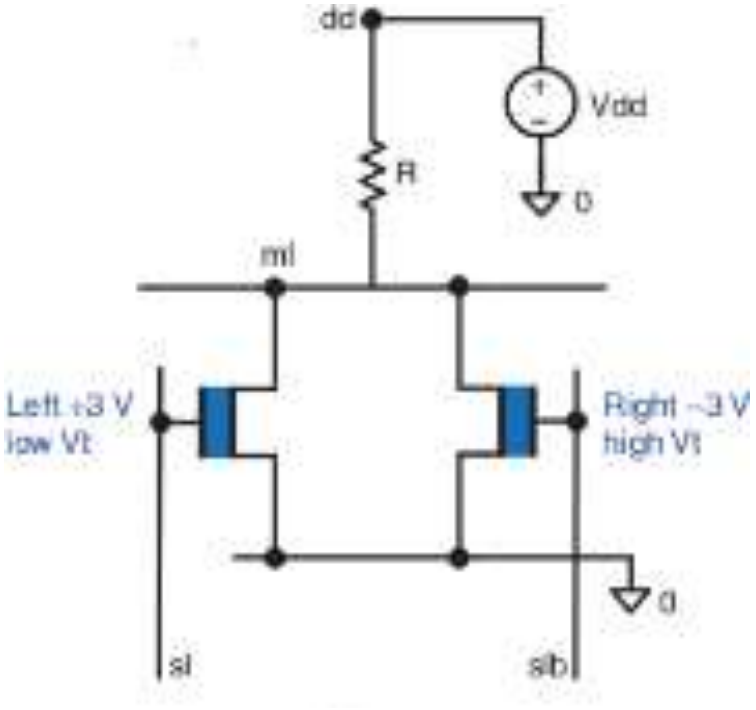


Figure: CAM Cell [5]

CAM Cell Operations (Write 0 and Write 1)

Write Logic '0':

- Left FeFET: +3 V pulse (sets to $Low - V_{th}$).
- Right FeFET: -3 V pulse (sets to $High - V_{th}$).
- This results in down polarization for the left FeFET (except at the gate edges) and up polarization for the right FeFET.

Write Logic '1':

- Left FeFET: -3 V pulse (sets to $High - V_{th}$).
- Right FeFET: +3 V pulse (sets to $Low - V_{th}$).
- Before the application of pulse inputs, ml stays high as both FeFETs are in the offstate. With the application of pulse inputs, the low-Vt FeFET is switched on, and ml discharges.

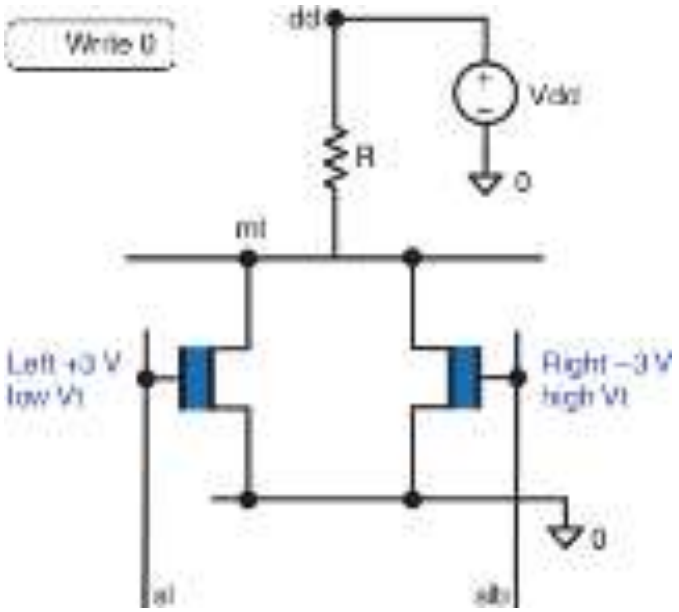


Figure: Write logic 0 [5]

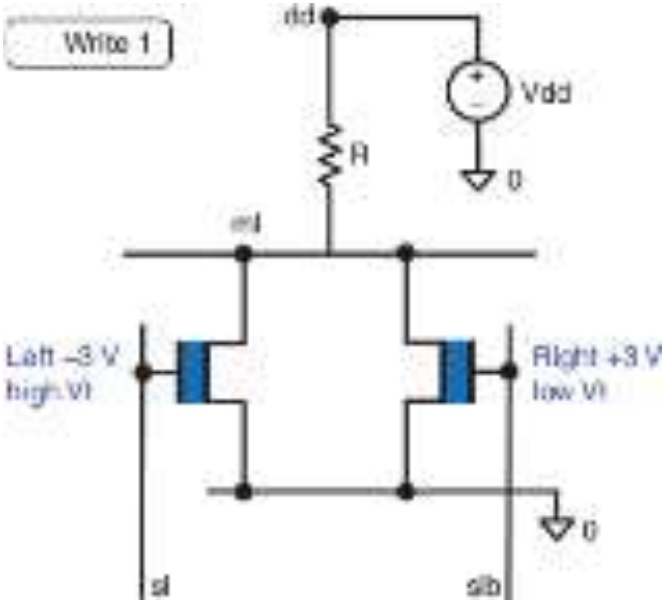


Figure: Write logic 1[5]

CAM Cell Simulation Result

Write Logic 0:

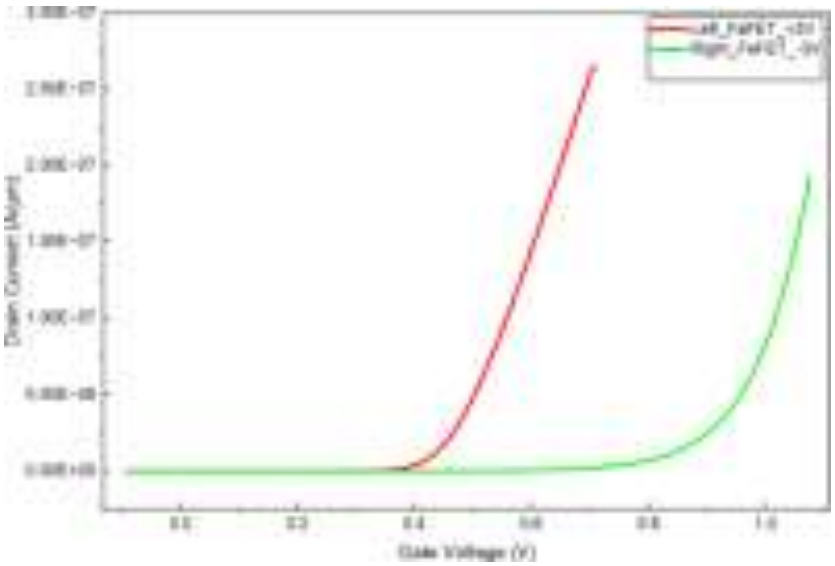


Figure: Ids vs Vgs linear scale post program state

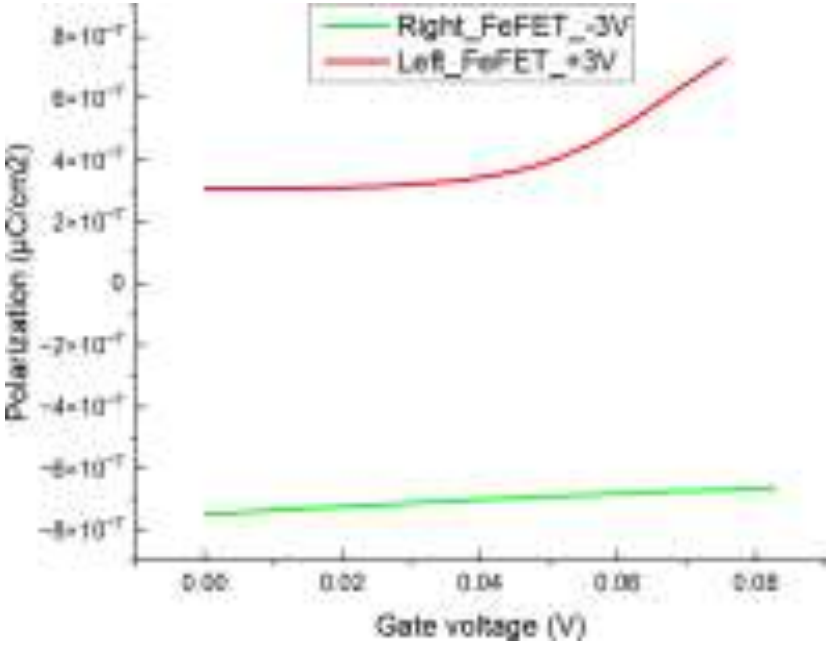


Figure: Id vs Vgs log scale post program state

CAM Cell Simulation Result

Write Logic 1:

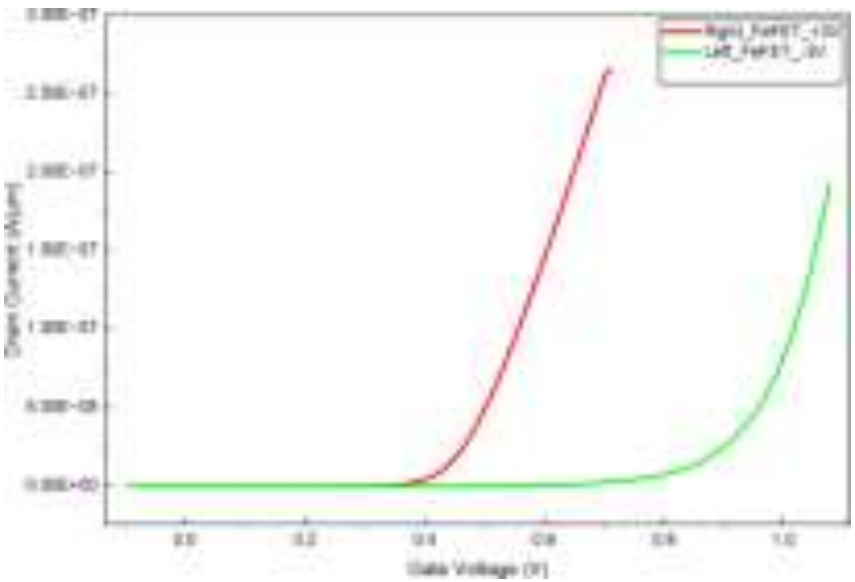


Figure: Ids vs Vgs linear scale post program state

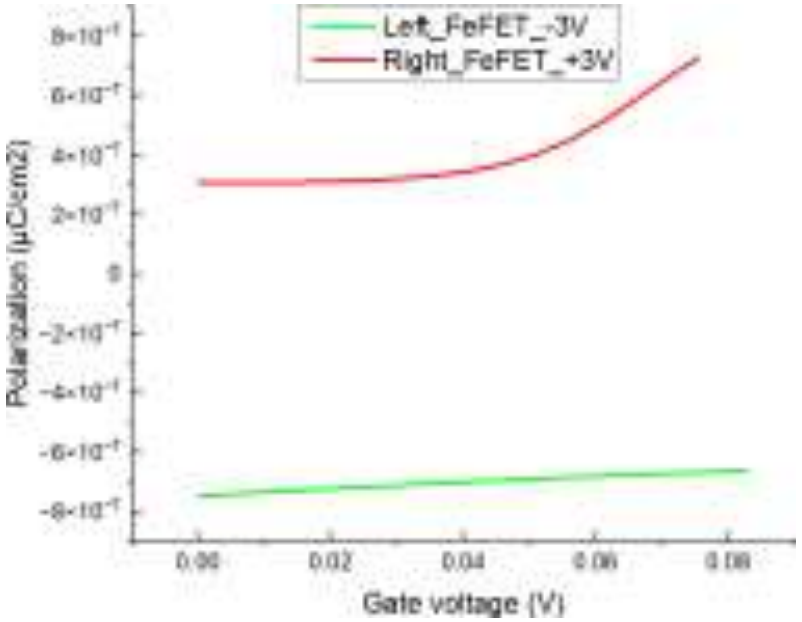


Figure: Id vs Vgs log scale post program state

Search Operation Mechanics

- **Search :**is used to perform massively parallel pattern matching in a single clock cycle.
- **Mechanism:** Apply search voltage $V_r = 0.6$ V to complementary search lines (SL/SLB).
- **Match Condition**: Input matches stored bits → Both FeFETs stay OFF → ML remains HIGH.
- **Mismatch Condition:** Input differs → One FeFET turns ON → ML discharges to ground (LOW).

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Future Work

Transition to HZO: Changing HfO_2 to $Hf_{0.5}Zr_{0.5}O_2$ (HZO). HZO has a lower crystallization temperature (400°C), improving compatibility with 2D MoS_2.

Proposed Architecture: Implementation of a Ferroelectric Embedded Dual-Gate (FEDG) FET, featuring a Top Gate for logic input and a Back Gate (with HZO) for memory storage.[4]

This configuration allows logic functions (e.g., AND/OR) to be implemented within a single transistor by combining the non-volatile memory state with active logic inputs.

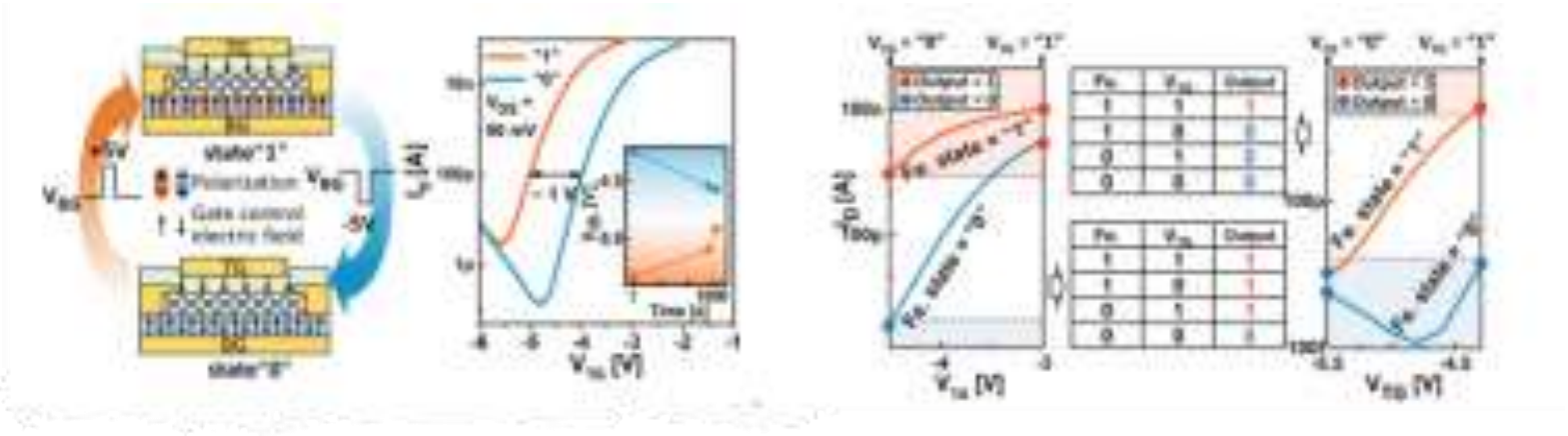

Figure: Left to Right:) Schematic drawing of the electrical field and polarization of the device in two different states,) Top-gate transfer characteristics of the device after different voltage pulse on the back-gate,Demonstration of FeFET with logic “and”, Demonstration of FeFET with logic “or”. operations[4]

References I

[1] Zheng, Y., Ravichandran, H., Schranghamer, T.F. et al. Hardware implementation of Bayesian network based on two-dimensional memtransistors. Nat Commun 13, 5578 (2022). https://doi.org/10.1038/s41467- 022-33053-x

[2] Wang, J.; Bi, J.; Xu, Y.; Niu, G.; Liu, M.; Stempitsky, V. Effects of Charge Trapping on Memory Characteristics for HfO2-Based Ferroelectric Field Effect Transistors. Nanomaterials 2023, 13, 638. https://doi.org/10.3390/nano13040638

[3]Pin-Chun Shen ,Chungwei Lin,Haozhe Wang, Koon Hoo Teo, Jing Kong"Ferroelectric memory field-effect transistors using CVD monolayer MoS2 as resistive switching channel".Appl. Phys. Lett. 116, 033501 (2020). https://doi.org/10.1063/1.5129963

[4] K. Huang et al., "Hf.Zr.O Ferroelectric Embedded Dual-Gate MoS Field Effect Transistors for Memory Merged Logic Applications," in IEEE Electron Device Letters, vol. 41, no. 10, pp. 1600-1603, Oct. 2020, doi: 10.1109/LED.2020.3019681. keywords: Molybdenum;Sulfur;Field effect transistors;Logic gates;Memory management;Fabrication;Electric fields;MoS;ferroelectric;HZO;dual-gate;memory merged logic;in memory computing